

Rheology



Learning objectives

At the end of this topic students should be able to

- ✓ Define rheology
- ✓ Understand the difference between newtonian and non- newtonian flow
- ✓ Describe each types of non-newtonian system
- ✓ Understand the determination of viscosity of fluids
- ✓ Describe the pharmaceutical application of rheology

Introduction

- Rheology is the branch of physics that deals with the *flow of liquids* and the *deformation of solids*.
- It is derived from, the Greek word *rheo* (“to flow”) and *logos* (“science”)

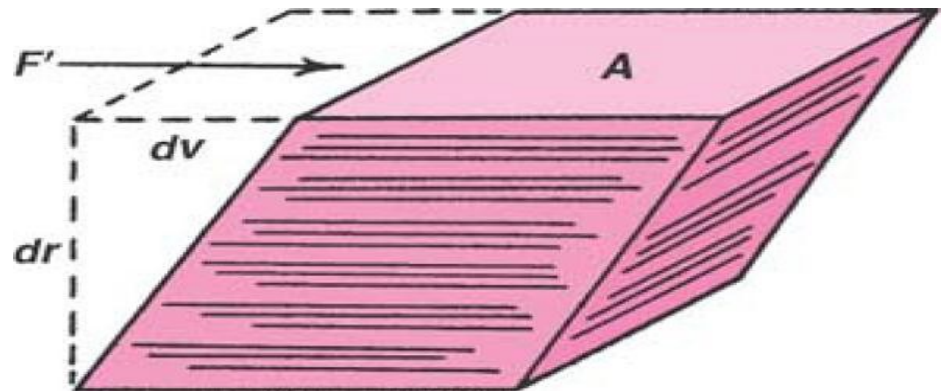
Introduction.....

Viscosity is an expression of the **resistance of a fluid to flow** (*or internal friction in fluid*).

- **The higher the viscosity, the greater is the resistance to flow.**
- Water, which is easier to stir than syrup, is said to have the lower viscosity.
- **The reciprocal of viscosity is fluidity.**

Introduction.....

- Consider a block of liquid consisting of parallel plates of molecules, similar to a deck of cards, as shown in Figure below
- If the bottom layer is fixed in place and the top plane of liquid is moved at a constant velocity, **each lower layer will move with a velocity directly proportional to its distance from the stationary bottom layer.**
- The difference of velocity, **dv** , between two planes of liquid separated by an infinitesimal distance, **dr** , is the **velocity gradient or rate of shear, dv/dr** .
- The force per unit area, F'/A , required to bring about flow is **called the shearing stress and is given the symbol F** .



Newtonian system

- **Newton** was the first to study flow properties of liquids in a quantitative way. **He recognized that the higher the viscosity of a liquid, the greater is the force per unit area (shearing stress) required to produce a certain rate of shear.**
- Rate of shear is given the symbol **G**. Hence, rate of shear should be directly proportional to shearing stress, or gradient between the parallel planes of a block of material.
- where **η** is the coefficient of viscosity, usually referred to simply as viscosity.

$$\frac{F'}{A} = \eta \frac{dv}{dr}$$

Equation1

$$\eta = \frac{F}{G}$$

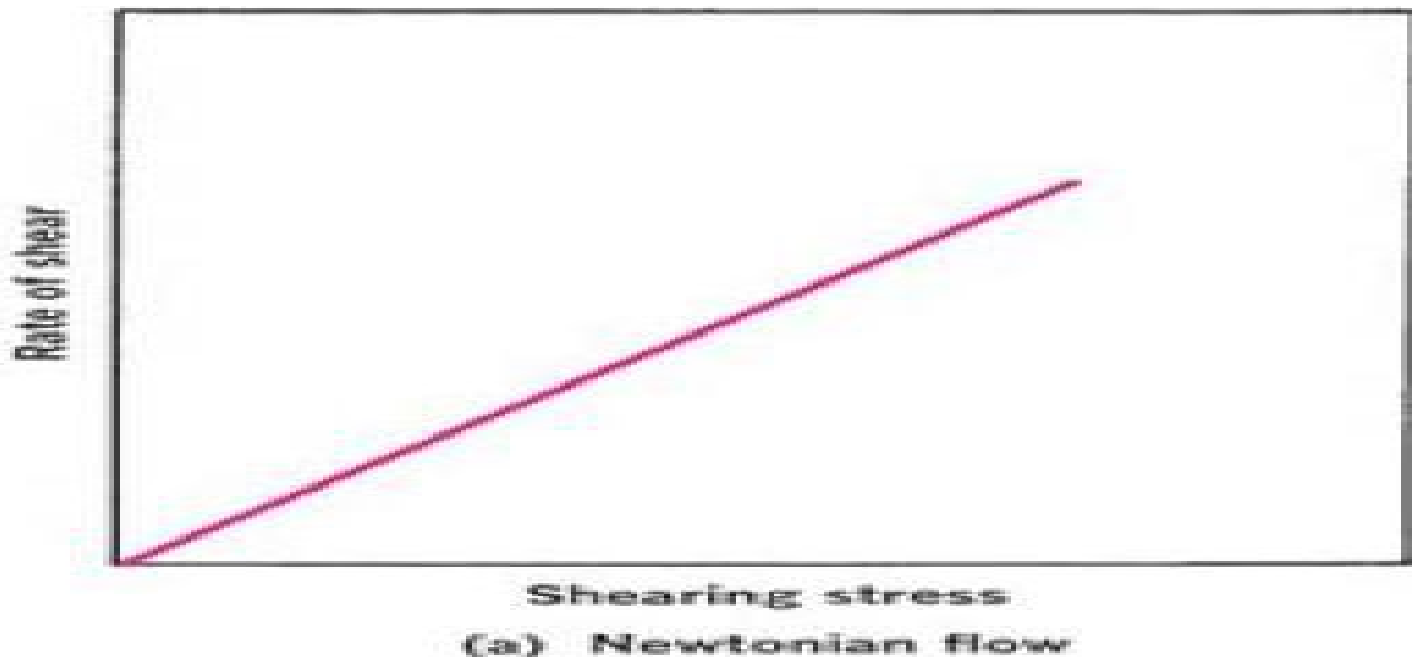
where $F = F' / A$ and $G = dv / dr$.

Newtonian system.....

- Based on the types of flow and deformation materials can be classified as one of the two categories:
 - Newtonian or
 - Non-Newtonian systems.
- The choice depends on whether or not their flow properties are in accordance with Newton's law of flow.

Newtonian system.....

- A representative **flow curve**, or **rheogram**, obtained by plotting F versus G for a Newtonian system.
- straight line passing through the origin is obtained.
- **Viscosity is the slope of such a line.**



Newtonian system

- **The unit of viscosity is the poise(p)**, as the shearing force required to produce a velocity of 1 cm/sec between two parallel planes of liquid each 1 cm² in area and separated by a distance of 1cm.

The cgs units for poise are g/cm sec.

Newtonian system

- These units are readily obtained by a dimensional analysis of the viscosity coefficient.
- Rearranging equation 1) to

$$\eta = \frac{F' dr}{A dv} = \frac{\text{dynes} \times \text{cm}}{\text{cm}^2 \times \text{cm/sec}} = \frac{\text{dyne sec}}{\text{cm}^2}$$

$$\frac{\text{dyne sec}}{\text{cm}^2} = \frac{\text{g} \times \text{cm/sec}^2 \times \text{sec}}{\text{cm}^2} = \frac{\text{g}}{\text{cm sec}}$$

Newtonian system

- Simple liquids follow Newton's law of direct proportionality between **shear stress** and **rate of shear**, so that their **viscosity** is **independent of the shear stress or the rate of shear**.

$$\eta = \frac{F}{G}$$

- Their flow behavior is thus referred to as **Newtonian**.
- Substances such as **water, ethanol, ethyl ether, glycerin** are **Newtonian fluids**.

Newtonian system

- The viscosity of simple liquids, i.e., **pure liquids consisting of small molecules and solutions where solute and solvent are small molecules**, depends only on **composition, temperature, and pressure**.
- For solutions of **solid solutes**, the viscosity usually **increases with concentration**.
- The **viscosity increases** moderately with **increasing pressure** and **markedly with decreasing temperature**.

Temperature Dependence of Viscosity

- Whereas the viscosity of a gas increases with temperature, that of a liquid decreases as **temperature is raised**, and the fluidity of a liquid (the reciprocal of viscosity) increases with temperature.
- The dependence of the viscosity of a liquid on temperature is expressed approximately for many substances by an **equation analogous to the Arrhenius equation of chemical kinetics**

Temperature Dependence of Viscosity

$$\ln \eta = \ln A + \frac{E_v}{RT}$$

- where **A** is a constant depending on the **molecular weight** and
- **E_v** is an activation energy required to initiate flow between molecules.
- **R** is a constant.-----8.314 J/K.mol
- **T** is temperature in °K

Temperature Dependence of Viscosity

- **More energy is required to break bonds and permit flow in liquids** composed of molecules that are associated through hydrogen bonds.
- **These bonds are broken at higher temperatures** by thermal movement, however, and η_v decreases markedly.

Non-Newtonian system

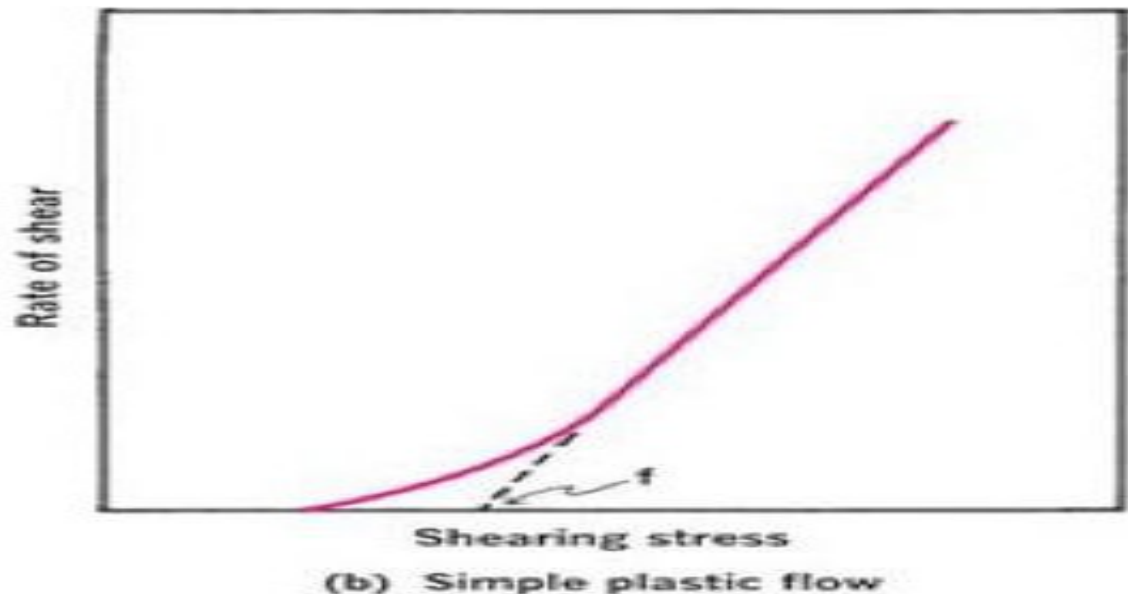
- Fluids that do not obey Newton's Law are often termed as **non-Newtonian fluids**.
- Non-Newtonian behaviour is generally exhibited by **colloidal dispersions, suspensions, emulsions, ointments, creams, gels, pastes, etc.**

Plastic flow

- ❑ Semisolids that do not flow at low shear stresses (exhibiting reversible deformation like elastic solids) but flow like liquids above their yield value (i.e., yield stress) are termed *plastic or Bingham bodies*.
- ❑ This type of rheological behavior is termed as *plasticity*.

Plastic flow.....

- Plastic body does not begin to flow until shearing stress corresponding to the yield value is exceeded.
- At shear stress below the yield value, the substance acts as an elastic material.



Plastic flow.....

- Plastic flow curves do not pass through the origin, but rather intersect the shearing stress axis (or will if the straight part of the curve is extrapolated to the axis) at a particular point referred to as **the yield value**.
- Plastic flow is exhibited by **concentrated suspensions**, particularly if the continuous phase is of high viscosity or if the particles are **flocculated**.

Plastic flow.....

- **A yield value** exists because of the contacts between adjacent particles (brought about by van der Waals forces), which must be broken down before flow can occur.
- **Frictional forces** between moving particles can also contribute to yield value.

Plastic flow...

- Once the yield value is exceeded, any further increase in shearing stress (i.e., $F - f_{\text{yield}}$) brings about a direct proportional increase in rate of shear.

$$U = \frac{F - f_{\text{yield}}}{G}$$

U is plastic viscosity, **F** is sheer stress

f_{yield} - is the yield value, or intercept, on the shear stress axis **G** is shear rate

- In effect a plastic system resembles a Newtonian system at shear stresses above the yield value.

Plastic flow...

Example.

- A plastic material was found to have a yield value of 5200 dynes/cm². At shearing stresses above the yield value, F was found to increase linearly with G . If the rate of shear was 150 sec⁻¹ when F was 8000 dynes/cm², calculate U , the plastic viscosity of the sample.

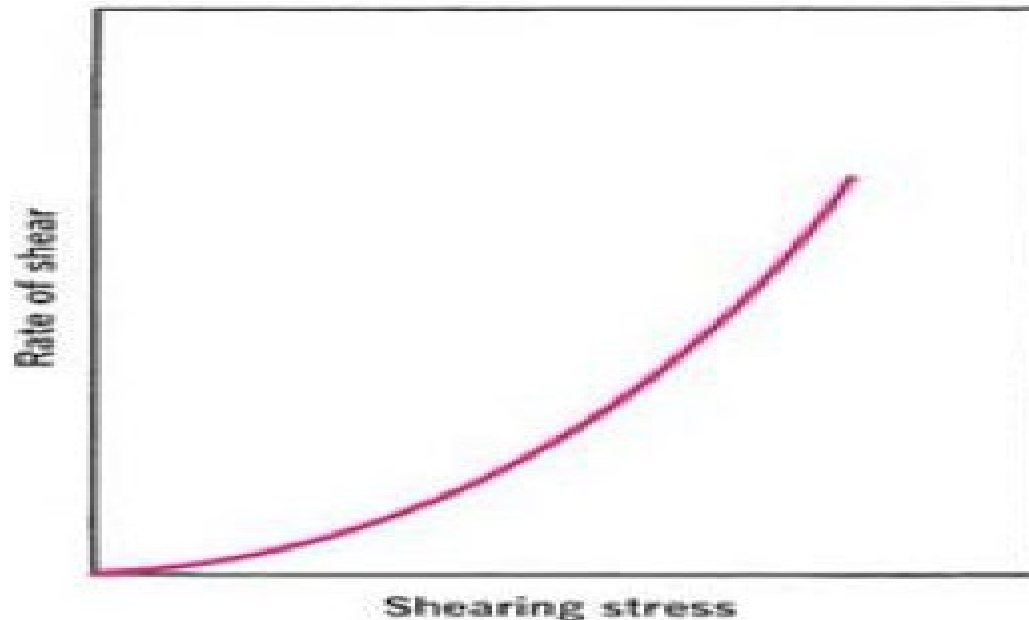
Ans. 18.67 poise

Pseudoplastic flow (shear thinning fluids)

- Many **polymer solutions** become more fluid the faster they are stirred.
- In shear thinning fluids, the viscosity decreases with increasing shear.

Pseudoplastic flow (shear thinning fluids)...

- As the **increase in shear rate is greater than the increase in the corresponding shear stress**, the flow curve is concaved toward the shear rate axis.



(c) Simple pseudoplastic flow

Pseudoplastic flow (shear thinning fluids).....

- When shear is applied, the randomly coiled, entangled macromolecules tend to disentangle themselves and align themselves in the direction of flow.

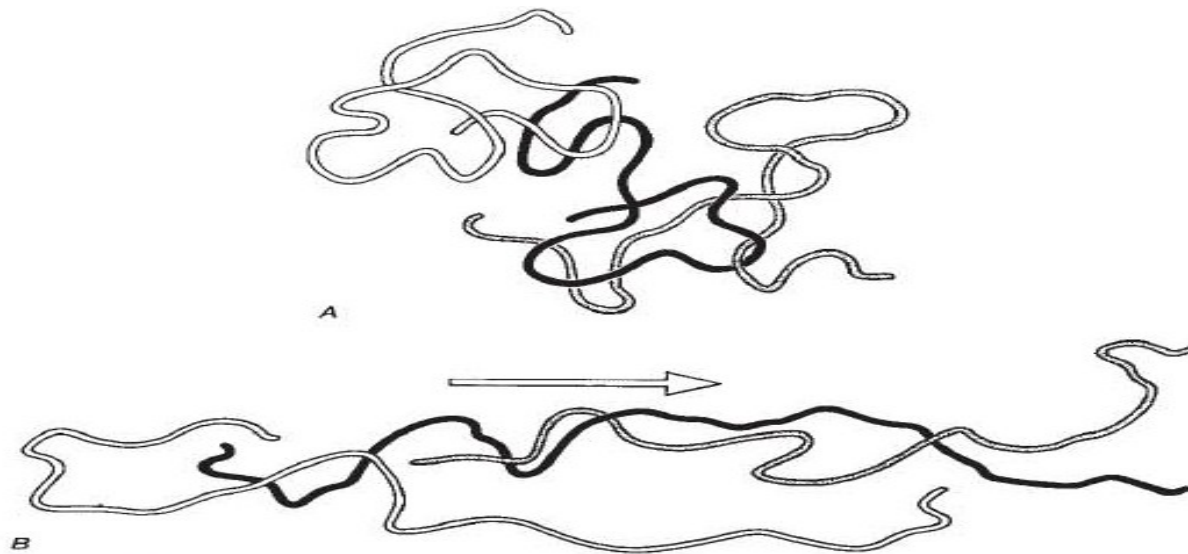


Figure 23-7. Three randomly coiled polymer chains in solution. **A.** At rest; **B.** In shear field.

Pseudoplastic flow (shear thinning fluids)....

- The imposition of increasing shear in these systems enables the macromolecule “chains” to uncoil progressively and become streamlined or elongated, thereby offering less resistance to flow than the original, approximately spherical shapes.

Pseudoplastic flow (shear thinning fluids)....

- There is an apparent viscosity for each value of shear rate or shear stress.

$$\eta' = \frac{F}{G^N} \quad \text{where } N \neq 1, \quad N < 1$$

$$\ln \eta' = N \ln F - \ln G$$

Pseudoplastic flow (shear thinning fluids)....

- At very high shear rates, the dissolved polymer chains are wholly disentangled and well align in the direction of flow, and the aggregates of particles are broken up as far as possible.
- There is no residual structure left which can be broken up by further increment in shear rate:
 - the viscosity levels off at a constant value called the upper Newtonian viscosity, η_{infinity} .

Pseudoplastic flow (shear thinning fluids).....

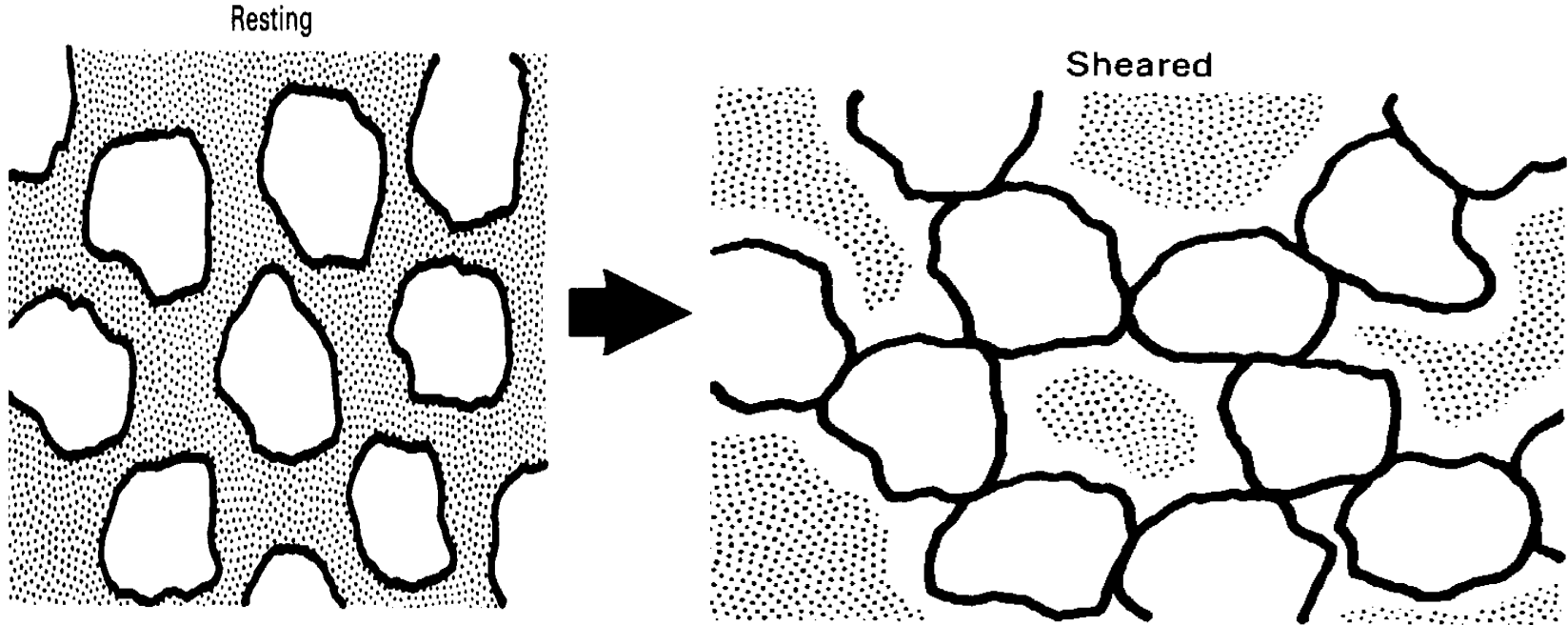
- Many pharmaceutical products, including liquid dispersions of natural and synthetic gums e.g., tragacanth, sodium alginate, methylcellulose and sodium carboxymethylcellulose exhibit pseudoplastic flow.

Dilatant flow (shear thickening)

- Certain suspensions with a **high percentage of dispersed solids** exhibit an **increase in viscosity with increase in shear**.
- When **concentrated suspension** is poured or stirred slowly, there is just enough liquid to lubricate the slipping of one particle past another, and the viscosity is low.

Dilatant flow (shear thickening).....

When stirred fast, the particles get into each other's way, block each other and bunch up rather than slipping past each other.



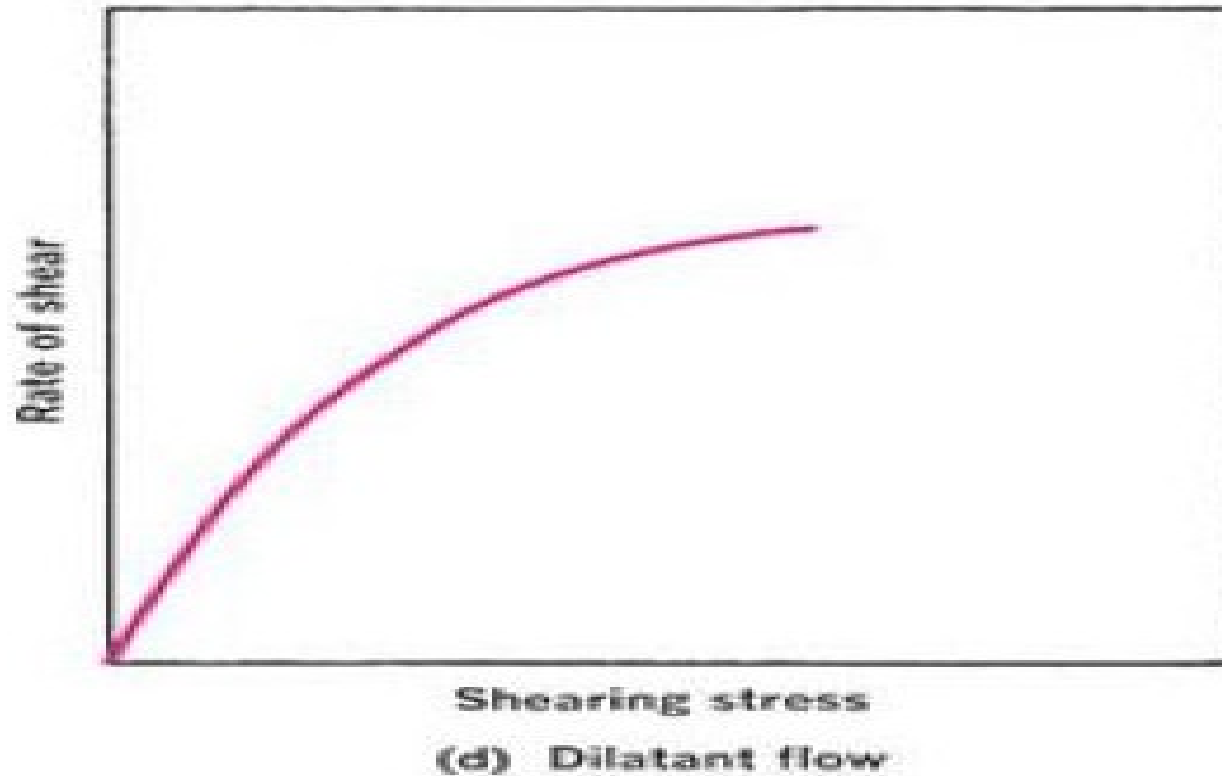
Dilatant flow.....

- Large voids form between the unevenly clustered particles, and as the **liquid seeps into these, the suspension appears dry** – as if the suspended solids had expanded or dilated.
- This phenomenon, which results in progressively viscosity increases, becomes more severe with increasing shear.

Dilatant flow.....

- When high shear is followed by low shear or rest, the particles that had been crowded together separate again, **the interparticle void volume decreases, suspension appears wet again and the viscosity drops.**
- Substances possessing dilatant flow properties are invariably **suspensions containing a high concentration (about 50% or greater) of small deflocculated particles.**

Dilatant flow.....



Thixotropy (Time dependent non-Newtonian flow)

- Substances whose viscosity depends on the **duration of shear** are said to be **thixotropic**.

For example, if a suspension is viscous and/or the particles are large and heavy, their Brownian motion is **too slow to restore the broken interparticle links instantaneously**.

Thixotropy.....

- Likewise, the entanglement of polymer chains are slow to be **re-established** by Brownian motion if their solution is **viscous**.

Thixotropy.....

- If the **rate of link restoration by Brownian motion is lower than the rate of link breakdown by shear, the apparent viscosity decreases even while the system is under constant shear**, as the number of the particles aggregates or the extent of macromolecular entanglement is progressively reduced.

Thixotropy.....

- Furthermore, the apparent viscosity at a given shear rate is lower if the system is stirred recently at high speed than if the shear rate was approached from low speed or from rest.

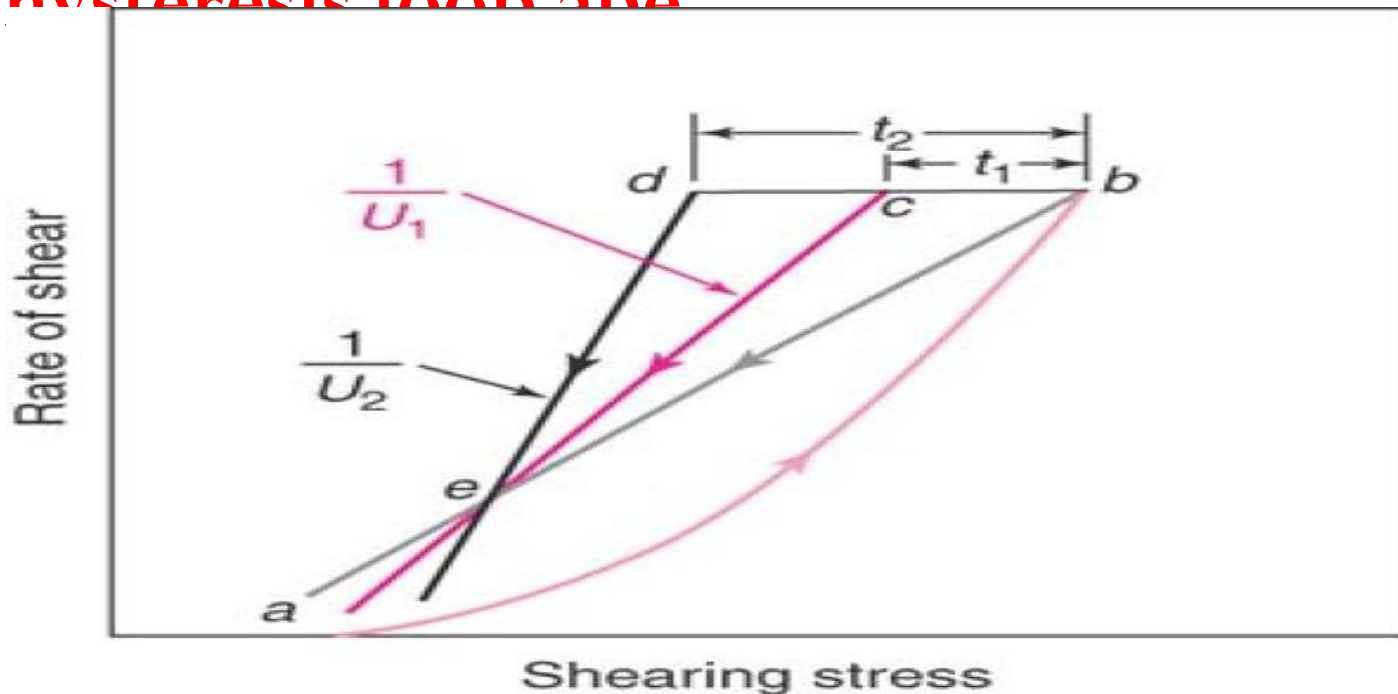
Thixotropy.....

- The apparent viscosity depends **not only on temperature, composition, rate of shear or shear stress, but on the previous shear history and time under shear.**

Thixotropy.....

- For example, suppose that in Figure below the shear rate of a thixotropic material is increased in a constant manner from point a to point b and is then decreased at the same rate back to e.

Typically, this would result in **the so-called hysteresis loop**



Thixotropy.....

- If, however, the sample was taken to point b and the shear rate held constant for a certain period of time (say, t_1 seconds), shearing stress, and hence consistency, would decrease to an extent depending on time of shear, rate of shear, and degree of structure in the sample.
- Decreasing the shear rate would then result in the hysteresis loop **abce**.
- If the sample had been held at the same rate of shear for t_2 seconds, the loop **abde** would have been observed.

Viscoelasticity

- When stressed, viscoelastic materials simultaneously exhibit some of the **properties of elastic solids** and some of the **properties of viscous liquids**:
- **some deformation occur instantaneously upon the application of stress and continuous as long as the stress is applied.**

Viscoelasticity...

- Upon the removal of the stress, there is **partial recovery of the original shape.**
- In effect, viscoelastic systems are capable of storing part of the deformation energy **elastically and reversibly.**
- E.g, Viscoelastic materials are molten polymers, chewing gums and most gels.

Determination of viscosity

Viscosity is measured by a viscometer.

- **A single point viscometer** can be used to determine viscosity of Newtonian fluids.
Eg .u shaped ostwald viscometer .
- **Multiple point viscometer** should be used to determine the viscosity **of non-newtonian fluids**.
Since the viscosity should be determined at different shear rate(rpm).
- Multiple point viscometer can also be used to measure Newtonian fluids.
 - Eg rotational viscometer.

Determination of viscosity.....

- **Dynamic viscosity(absolute viscosity)** is a measure of internal resistance of a fluid. Gives information on the force needed to make the fluid flow at certain rate. Represented by the symbol η
- **Kinematic viscosity** is a measure of a fluid's internal resistance to flow under gravitational force. It tells you how fast the fluid is moving when a certain force is applied.
- It is determined by measuring the time in seconds, required for a fixed volume of fluid to flow a known distance by gravity through a capillary within a calibrated viscometer at a closed controlled temperature.
- **Kinematic viscosity is the absolute viscosity(dynamic viscosity) divided by the density of the liquid at a specific temperature.** The units of kinematic viscosity are the stoke (s) and the centistokes(cs).

$$\text{Kinematic viscosity} = \frac{\eta}{\rho}$$

Where η the absolute viscosity and ρ is the density.

Determination of viscosity

OSTWALD VISCOMETER:- I

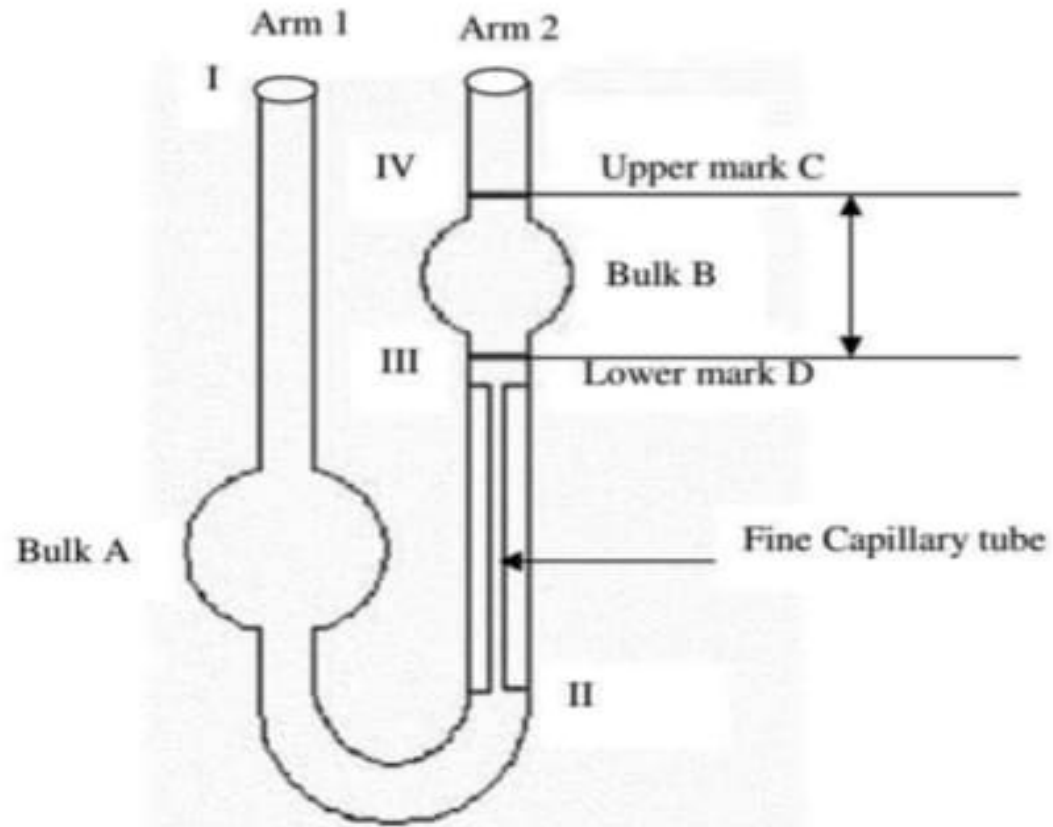


Fig. 1:- Ostwald Viscometer

Determination of viscosity

The Kinematic viscosity of liquid can be determined by the following formula.

$$\frac{\eta_1}{\eta_2} = \frac{\rho_1 t_1}{\rho_2 t_2}$$

- The value $\eta_1/\eta_2 = \eta_{rel}$ is known as the relative viscosity of the liquid under test.
- $\frac{\rho_1}{\rho_2}$ Is the density of liquid 1
- ρ_2 Is the density of liquid 2
- t_1 is the time taken for liquid 1 to flow from upper mark to the lower mark
- t_2 is the time taken for liquid 2 to flow from upper mark to the lower mark.

Determination of viscosity

Example.

- The time required for acetone to flow between the two marks on the capillary viscometer was 45 sec and for water the time was 100 sec, at 25°C. The density of acetone is 0.786 g/cm³ and that of water is 0.997 g/cm³ at 25°C. The viscosity of water is 0.8904 cp at this temperature. Find the viscosity of acetone at 25°C.
- **Ans. 0.316 cp.**

Pharmaceutical applications of rheology

1. Fluids

- ✓ Mixing
- ✓ Particle size reduction of disperse systems with shear.
- ✓ Passage through orifices, including pouring, package in bottles, and passage through hypodermic needles.
- ✓ Fluid transfer, including pumping and flow through pumps.
- ✓ Physical stability of disperse systems.

Pharmaceutical applications of rheology...

2. Semisolids

- ✓ Spreading and adherence on the skin
- ✓ Removals from jars or extrusion from tubes
- ✓ Release of drug from base

3. Solids

- ✓ Flow of powders from hoppers and into die cavities in tableting or into capsules during encapsulation
- ✓ Packagability of powdered or granular solids

4. Processing

- Production capacity of the equipment
- Processing efficiency

????